
Network Traffic Features

Understanding Connection Attributes for Network Analysis

Key features used in network traffic analysis and intrusion detection

1 Timestamp

What it is : The date and time when the connection started or was logged.

Example : 2024-03-15 14:23:05

Why it matters : Allows time-based analysis, such as detecting attacks that occur at specific hours (e.g., nighttime scans) or correlating events across multiple logs.

2 Duration

What it is : The length of the connection, in seconds.

Example : 24.0777 → this connection lasted about 24 seconds.

Why it matters : Very short or very long durations can indicate unusual activity.

- Extremely long connection with lots of packets may be suspicious.

3 Protocol

What it is : The communication protocol used for this connection.

Examples :

- TCP → reliable connection-oriented protocol
- UDP → fast, connectionless protocol
- ICMP → used for ping/testing network

Why it matters : Some attacks use specific protocols (like ICMP floods or TCP scans).

4 SourceIP

What it is : The IP address of the machine that **initiates the connection**.

Example : 192.168.1.239

Why it matters : Helps identify which machine is communicating or behaving suspiciously.

5 DestinationIP

What it is : The IP address of the machine **receiving the connection**.

Example : 192.168.1.234

Why it matters : You can see which targets are being contacted, and build the network graph (nodes = IPs).

6 SourcePort

What it is : The port number on the **source machine** used for the connection.

Example : 8055

Why it matters : Certain ports are used for specific services; abnormal port activity may indicate scanning or attacks.

7 DestinationPort

What it is : The port number on the **destination machine** being accessed.

Example : 1

Why it matters : Attacks may target specific ports (e.g., SSH = 22, HTTP = 80).

8 PacketCount

What it is : The number of **network packets** sent in this connection.

Example : 827 → 827 packets were transmitted.

Why it matters : High packet count in short duration can indicate flooding or attacks.

9 BytesSent

What it is : The total amount of **data sent from the source** to the destination, in bytes.

Example : 142500 → about 142 KB were uploaded.

Why it matters : High bytes sent relative to bytes received may indicate data exfiltration or upload-heavy attacks.

10 BytesReceived

What it is : The total amount of **data received from the destination**, in bytes.

Example : 55744 → about 55.7 KB were downloaded.

Why it matters : Useful for detecting download-heavy activity (e.g., malware retrieval, data theft).

11 Label

What it is : Classification of the connection.

Examples :

- **Normal** → normal traffic
- **Attack** → suspicious or malicious activity

Why it matters : Useful for training machine learning models or testing your IDS.

☰ Summary

Feature	Description
Timestamp	when the connection occurred
Duration	how long it lasted
Protocol	type of network communication
SourceIP / DestinationIP	who is talking to whom
SourcePort / DestinationPort	which service/port is used
PacketCount	number of packets sent
BytesSent / BytesReceived	data uploaded vs downloaded
Label	whether it's normal or an attack

TABLE 1 – Summary of Network Traffic Features

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